

Dengue Outbreak Prediction Using Machine Learning and Weather Data

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Dengue fever remains a major public health challenge in many tropical countries, including Sri Lanka, where frequent outbreaks place a significant burden on healthcare systems. The Colombo district consistently records one of the highest numbers of dengue cases due to its high population density, urbanization, and favorable climatic conditions for mosquito breeding. Despite the availability of surveillance systems, many existing dengue prediction models function as black-box systems that provide predictions without clear explanations or user-friendly analytical tools for public health officials.

This research aims to develop an advanced analytics and explainable dashboard system that transforms dengue prediction outputs into interpretable and actionable insights. The proposed solution integrates machine learning-based dengue risk prediction with interactive data visualization and Explainable Artificial Intelligence (XAI) techniques. SHAP (SHapley Additive Explanations) is used to explain the influence of different environmental and climatic factors on dengue predictions. The system also incorporates spatial analysis using GIS-based visualization methods to identify high-risk locations within the Colombo district.

The proposed platform will utilize technologies such as Python for data analysis and machine learning, GIS mapping tools for spatial visualization, and modern dashboard frameworks for interactive visualization. The dashboard will present temporal trends, climate-dengue correlations, risk heatmaps, and model explanation outputs that can assist decision makers in understanding and interpreting dengue outbreak predictions.

The expected outcome of this research is an intelligent analytical dashboard that enhances transparency, usability, and practical applicability of dengue prediction systems. By transforming complex predictive outputs into understandable visual insights, the system can support public health authorities in early outbreak detection, resource planning, and targeted intervention strategies. Ultimately, this research contributes to improving data-driven decision making in dengue surveillance and prevention.

Keywords: Dengue prediction, Explainable AI, Data analytics dashboard, Spatial risk analysis, Public health surveillance.

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LIST OF ABBREVIATIONS

Abbreviations	Descriptions
AI	Artificial Intelligence
API	Application Programming Interface
ARO	Annual Rate of Occurrence
AV	Asset Value
CSV	Comma Separated Values
DSS	Decision Support System
GIS	Geographic Information System
ML	Machine Learning
SHAP	SHapley Additive Explanations
WHO	World Health Organization
XAI	Explainable Artificial Intelligence

1. INTRODUCTION

1.1 Background and Context

Dengue fever is one of the most significant mosquito-borne viral diseases affecting tropical and subtropical regions worldwide. The disease is primarily transmitted by the *Aedes aegypti* mosquito and continues to pose a serious public health threat, particularly in densely populated urban areas. According to the World Health Organization (WHO), dengue infections have increased dramatically over the past few decades, with nearly half of the global population now at risk of infection. Countries in South and Southeast Asia report a large proportion of global dengue cases due to favorable climatic conditions that support mosquito breeding and transmission.

Sri Lanka has experienced recurring dengue outbreaks for many years, with the Western Province consistently reporting the highest number of cases. Among these regions, the Colombo district has been identified as one of the most affected areas due to its high population density, rapid urbanization, and environmental conditions that facilitate mosquito breeding. Seasonal rainfall patterns, temperature fluctuations, and humidity levels significantly influence dengue transmission patterns in the region. As a result, early identification of potential outbreaks is essential to support public health authorities in implementing timely prevention and control measures.

Currently, dengue surveillance systems mainly rely on historical case reporting and basic statistical analysis to monitor disease trends. While several machine learning-based prediction models have been proposed in recent years, many of these systems operate as black-box models that provide predictions without explaining the factors influencing those predictions. This lack of interpretability makes it difficult for healthcare decision-makers to fully trust and utilize these systems in real-world decision-making. Furthermore, many existing tools lack advanced data visualization and spatial analysis capabilities that can help public health officials understand disease patterns across time and geographic locations.

To address these limitations, this research proposes the development of an advanced analytics and visualization platform that integrates machine learning predictions with explainable artificial intelligence (XAI) and interactive dashboard technologies. The system will focus specifically on dengue surveillance within the Colombo district and will transform complex analytical outputs into interpretable and actionable insights for public health authorities. By integrating temporal analysis, climate-dengue correlation analysis, and GIS-based spatial

visualization, the proposed solution aims to enhance the effectiveness of dengue monitoring and outbreak prediction.

This research aligns with the United Nations Sustainable Development Goal (SDG) 3: Good Health and Well-Being, which aims to ensure healthy lives and promote well-being for all. Additionally, the study contributes to the research cluster of Data Science and Artificial Intelligence by applying advanced analytics and explainable machine learning techniques to address a real-world public health challenge.

1.2 Literature Review

Recent studies have explored the use of machine learning and data analytics techniques to predict dengue outbreaks and analyze disease patterns. Climate variables such as rainfall, temperature, and humidity have been widely used to develop predictive models for dengue risk forecasting [1]. Environmental factors significantly influence mosquito population dynamics and dengue transmission patterns, particularly in tropical regions [2]. Several researchers have applied machine learning algorithms such as Random Forest, Support Vector Machines, and Artificial Neural Networks to identify complex relationships between climatic variables and dengue incidence [3]. These approaches have demonstrated improved prediction accuracy compared to traditional statistical methods [4].

Previous research also highlights that combining epidemiological data with meteorological variables significantly improves the reliability of dengue prediction models [5]. Such data-driven approaches allow researchers and public health authorities to analyze historical disease patterns and identify potential outbreak periods in advance. Studies have also shown that integrating environmental variables with dengue surveillance data enhances the predictive capability of outbreak forecasting systems [6].

In addition to predictive modeling, visualization techniques and geographic analysis have been increasingly applied in dengue surveillance systems. Geographic Information Systems (GIS) enable researchers to visualize the spatial distribution of dengue cases and identify geographic clusters where outbreaks are more likely to occur [1]. Spatial analysis methods also help public health authorities understand how dengue spreads across regions and how environmental conditions influence transmission patterns [2].

However, despite these advancements, many existing dengue prediction systems still suffer from several limitations. One major limitation is that many machine learning models operate as black-box systems, meaning that they generate predictions without clearly explaining the

factors contributing to those predictions. This lack of transparency reduces the trust of healthcare professionals and policymakers when using such models for public health decision-making [7].

To address this issue, Explainable Artificial Intelligence (XAI) techniques have been introduced to improve the interpretability of machine learning models. One widely used approach is SHAP (SHapley Additive Explanations), which explains how individual variables contribute to prediction results [8]. By providing insights into the importance of different variables, explainable models help researchers better understand the factors influencing dengue outbreak predictions.

Furthermore, recent studies emphasize the importance of integrating predictive models with interactive visualization platforms. Analytical dashboards that combine temporal analysis, spatial mapping, and explainable machine learning outputs can significantly enhance decision-making capabilities for health authorities [9]. Such systems allow policymakers to explore disease trends, identify high-risk areas, and interpret prediction results more effectively. These approaches are increasingly recommended in global dengue surveillance strategies [10].

1.3 Research Gap

Although numerous studies have applied machine learning techniques to dengue prediction, several important gaps remain in the existing research. Many current systems primarily focus on generating predictions without providing sufficient interpretability or explanation of the results. As a result, healthcare professionals may find it difficult to understand the underlying factors influencing dengue outbreaks, limiting the practical applicability of these predictive models.

Additionally, existing dengue surveillance systems often lack integrated analytical dashboards that combine temporal trends, environmental correlations, and spatial risk visualization within a single platform. This fragmentation makes it challenging for public health decision-makers to gain a comprehensive understanding of dengue patterns across different locations and time periods. Moreover, limited research has focused specifically on localized dengue surveillance solutions tailored to high-risk regions such as the Colombo district in Sri Lanka.

To address these gaps, this research proposes an explainable analytics platform that integrates machine learning prediction, SHAP-based explainability, and interactive visualization dashboards. By combining predictive modeling with transparent explanation methods and GIS-

based spatial analysis, the proposed system aims to provide interpretable and actionable insights for dengue surveillance. This approach enhances both the research novelty and the practical usefulness of dengue prediction systems, ultimately supporting more informed public health decision-making and targeted intervention strategies.

2. OBJECTIVE

2.1 Main Objective

To design and develop an interactive data analytics and visualization dashboard that integrates machine learning predictions and explainable artificial intelligence (XAI) techniques to analyze dengue outbreak patterns in the Colombo district and support public health decision-making.

2.2 Specific Objectives

1. To collect and preprocess dengue surveillance data and relevant climate data for the Colombo district to support predictive analysis.
2. To develop a machine learning model that predicts dengue outbreak trends using environmental and epidemiological data.
3. To apply Explainable Artificial Intelligence techniques such as SHAP to interpret the predictions generated by the machine learning model.
4. To design and implement an interactive dashboard that visualizes dengue trends, climate correlations, and predictive insights.
5. To perform spatial analysis and generate GIS-based risk maps to identify high-risk dengue areas within the Colombo district.
6. To evaluate the effectiveness of the proposed dashboard in providing interpretable and actionable insights for dengue surveillance and public health decision-making.

3. METHODOLOGY

3.1 Research Approach

This research adopts a Design Science Research (DSR) approach to develop and evaluate an analytical dashboard for dengue outbreak monitoring and prediction. Design Science is appropriate for this study because the research focuses on designing and implementing a technological artifact that addresses a real-world public health problem. The proposed artifact is an interactive data analytics platform that integrates machine learning predictions,

explainable artificial intelligence (XAI), and visualization tools to support dengue surveillance in the Colombo district.

The research process involves several stages including data collection, data preprocessing, predictive model development, explainable AI analysis, and dashboard implementation. The final system will be evaluated based on its ability to provide interpretable insights and support decision-making for dengue surveillance.

3.2 Technical Methods and Algorithms

The system will apply machine learning techniques to analyze historical dengue case data and environmental variables. These variables include climate factors such as rainfall, temperature, and humidity, which are known to influence mosquito breeding and dengue transmission patterns.

Machine learning algorithms such as Random Forest and Gradient Boosting models will be used to identify patterns and predict dengue outbreak risks. These models are chosen because of their strong performance in handling complex nonlinear relationships in epidemiological datasets.

To improve model transparency, Explainable Artificial Intelligence (XAI) techniques will be applied. SHAP (SHapley Additive Explanations) will be used to interpret the predictions generated by the machine learning model. SHAP provides insights into how each feature contributes to the final prediction, enabling researchers and public health officials to understand the factors influencing dengue outbreak predictions.

3.3 Tools and Platforms with Justification

Several software tools and technologies will be used to implement the proposed system.

Python will be used for data analysis and machine learning model development due to its extensive libraries such as **Pandas, NumPy, Scikit-learn, and SHAP**. These libraries support efficient data preprocessing, model training, and explainability analysis.

A **MySQL database** will be used to store dengue case records and climate-related data. This database will enable efficient data management, retrieval, and integration for predictive analysis.

For data visualization and dashboard implementation, a **web-based dashboard framework such as Flask or Dash** will be used to create interactive analytical dashboards. These

dashboards will display dengue trends, climate correlations, predictive insights, and feature importance explanations generated by the machine learning model.

To improve accessibility for public health authorities, a **mobile application interface will be developed using Flutter**, a cross-platform mobile development framework. The mobile application will allow users to access key dashboard insights, outbreak alerts, and dengue risk maps directly from mobile devices, enabling faster decision-making and real-time monitoring.

For spatial analysis and geographic visualization, **Geographic Information System (GIS) tools such as Mapbox or Leaflet** will be used to generate heatmaps and visualize dengue risk patterns across different areas of the Colombo district.

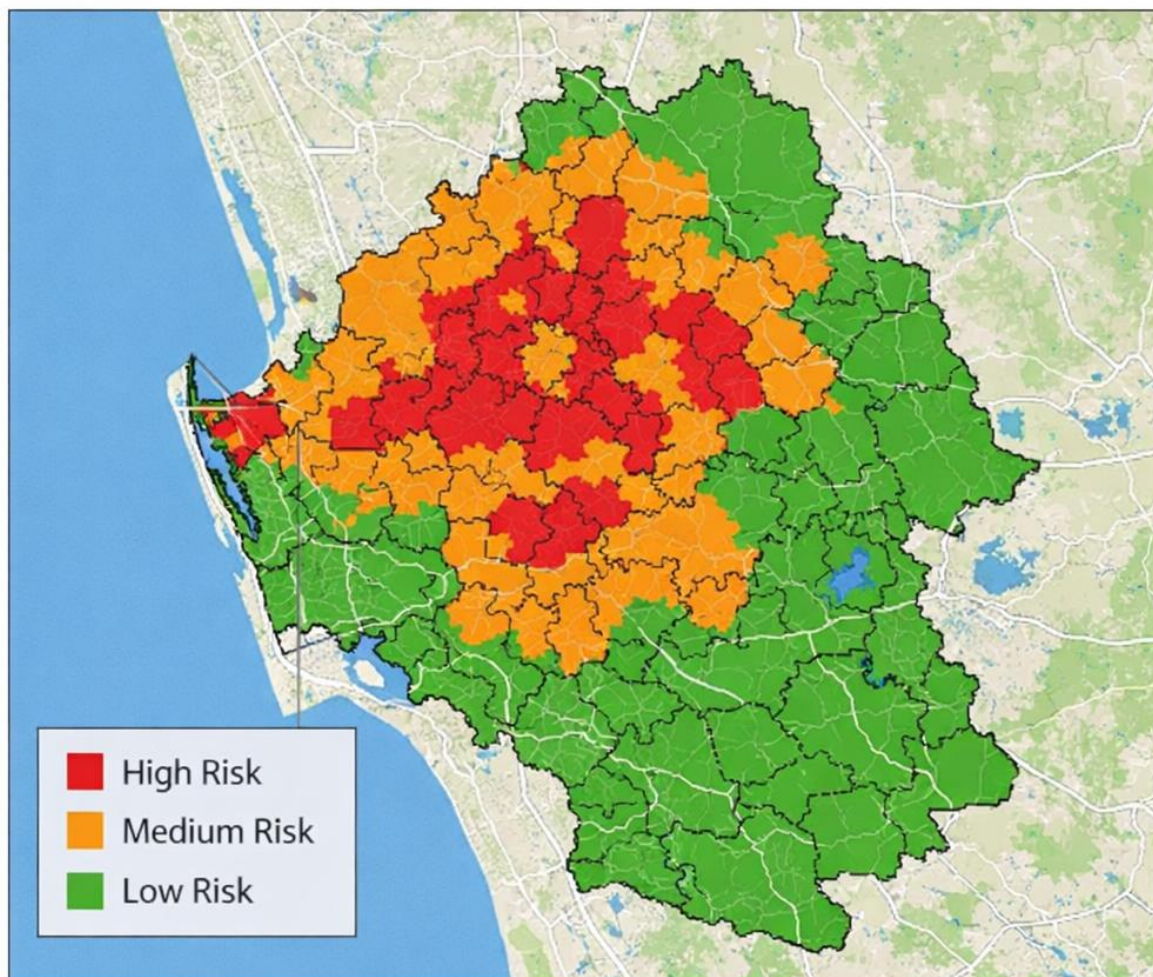


Figure 7: GIS-Based Dengue Risk Heatmap for Colombo District

3.4 Data Collection and Ethical Considerations

The Research will use dengue surveillance data and climate data related to the **Colombo district in Sri Lanka**. Dengue case data will be obtained from the **National Dengue Control Unit (NDU) of Sri Lanka**, which maintains official records of reported dengue cases across different districts. These datasets provide historical information about dengue incidence, which is essential for analyzing outbreak patterns and training predictive models.

Climate-related data such as **rainfall, temperature, and humidity** will be collected from the **Department of Meteorology of Sri Lanka**. Meteorological factors play an important role in mosquito breeding and dengue transmission; therefore, integrating climate data with dengue surveillance data will support the development of more accurate predictive models.

The collected datasets will undergo preprocessing steps such as data cleaning, missing value handling, and normalization before being used for machine learning analysis. These preprocessing steps ensure the quality and reliability of the data used in the predictive modeling process.

Since the research utilizes aggregated public health and climate datasets rather than individual patient records, the risk of privacy violations is minimal. Nevertheless, ethical considerations will be strictly followed by ensuring that no personally identifiable information is used during analysis. The data will be used solely for academic research purposes, and appropriate acknowledgment will be given to the data sources.

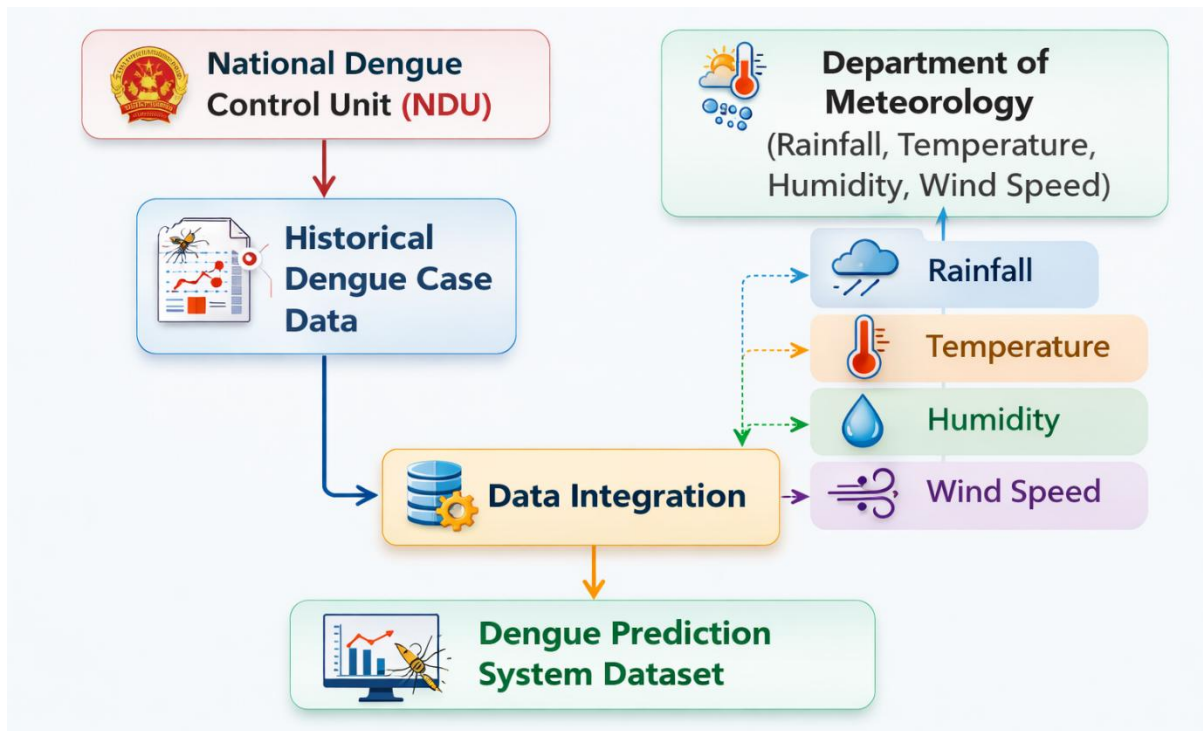


Figure 8 : Data Sources Used for Dengue Prediction System

Data Source	Organization	Data Type	Description
Dengue Surveillance Data	National Dengue Control Unit (NDU), Sri Lanka	Epidemiological Data	Historical dengue case records including date, location, and number of cases
Meteorological Data	Department of Meteorology, Sri Lanka	Climate Data	Weather data such as rainfall, temperature, humidity, and wind speed

Table 3: Data Sources Used for the Dengue Prediction System

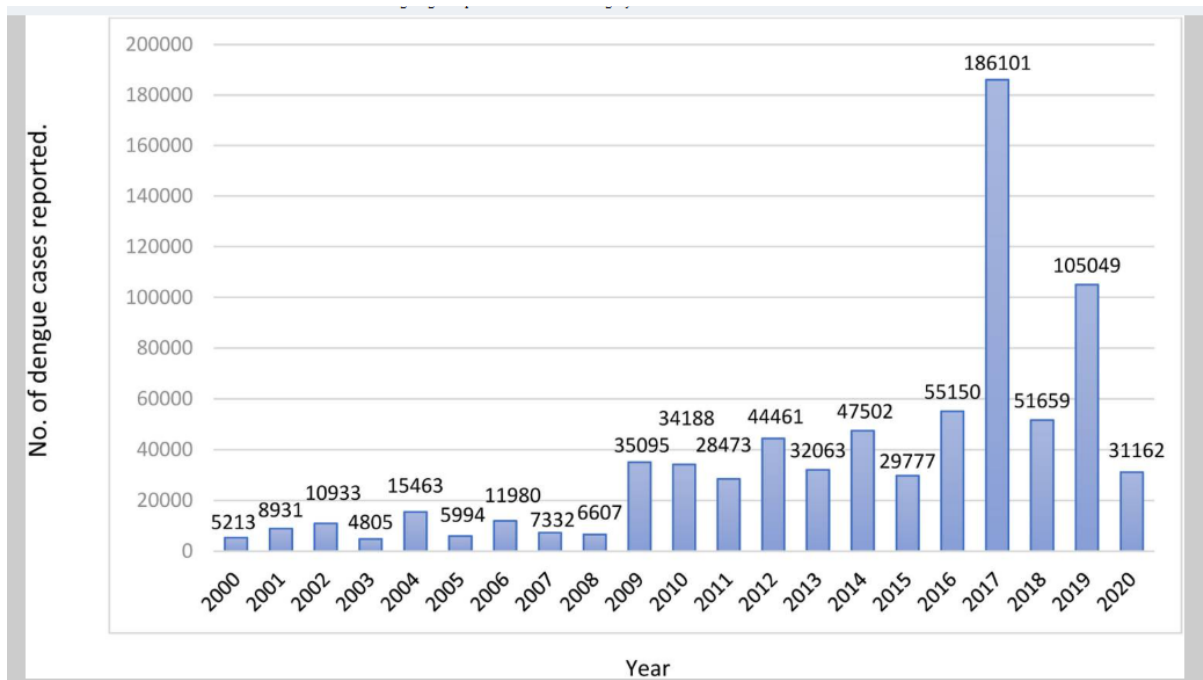


Figure 9 : Temporal Trend of Dengue Cases [11]

Validation Strategy and Performance Metrics

The predictive performance of the machine learning models will be evaluated using standard validation techniques. The dataset will be divided into **training and testing datasets** to ensure that the model can generalize to unseen data.

Several performance metrics will be used to evaluate the prediction accuracy of the model. These include **Accuracy, Precision, Recall, and F1-score**. These metrics help assess how well the model identifies dengue outbreak patterns.

In addition to predictive performance, the usability and interpretability of the dashboard will also be evaluated by assessing how effectively the visualization tools communicate insights about dengue trends, environmental correlations, and spatial risk patterns.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution and its Components

The proposed system is a dengue surveillance and analytics platform designed to support public health decision-making in the Colombo district. The system integrates multiple components including data collection, machine learning-based prediction, explainable artificial intelligence (XAI), and an interactive analytics dashboard.

Dengue case data will be obtained from the **National Dengue Control Unit (NDU) of Sri Lanka**, while climate data such as rainfall and temperature will be obtained from the **Department of Meteorology in Sri Lanka**. These datasets will be used by the system to generate predictive insights related to dengue outbreak risks.

The machine learning prediction model and explainability components will generate analytical outputs that indicate potential dengue risk levels and the contributing environmental factors. These outputs will then be integrated into an **interactive data analytics dashboard**, which represents the main component developed in this research.

The dashboard provides visual representations of dengue trends, spatial risk patterns, and environmental correlations, allowing public health authorities to interpret predictive results and make informed decisions regarding dengue prevention and control strategies.

4.2 System Diagram of Components

The overall system consists of several integrated components:

1. Data Sources

- Dengue surveillance data from the **National Dengue Control Unit (NDU)**
- Climate data from the **Department of Meteorology in Sri Lanka**

2. Data Processing and Prediction Module

This module processes collected data and generated dengue outbreak predictions using machine learning techniques. The module also produces explainability outputs using methods such as SHAP to identify the factors influencing predictions.

3. Data Analytics and Visualization Dashboard

The **data analytics dashboard** represents the primary contribution of this research. This component integrates predictive outputs generated by the machine learning module and transforms them into interpretable and actionable insights.

The dashboard enables users to:

- Visualize dengue case trends over time
- Analyse correlations between climate variables and dengue cases
- View SHAP-based feature importance explanations
- Explore GIS-based dengue risk maps across the Colombo district
- Filter and compare data by time period or geographic location

By transforming complex analytical outputs into intuitive visualizations, the dashboard improves the usability of dengue prediction systems for public health authorities.

4.3 Highlighting the Individual Component

This research primarily focuses on the data visualization and explainable analytics component, which integrates machine learning outputs with interactive dashboards. This component transforms complex predictive results into interpretable insights that can be easily understood by public health authorities.

The visualization component includes features such as temporal trend analysis, climate–dengue correlation visualizations, geographic heatmaps, and interactive filters that allow users to explore dengue risk patterns across districts and MOH areas.

4.4 Component Interactions

The system begins by collecting dengue case data and climate data from official sources. These datasets are processed and analyzed within the prediction module to generate dengue outbreak predictions and explainability results.

The generated analytical outputs are then transmitted to the **data analytics dashboard**, where they are visualized using charts, maps, and interactive visual elements. Users can interact with these visualizations to explore dengue trends, identify high-risk areas, and understand the key factors influencing outbreak predictions.

4.5 Data Flow Explanation

The data flow of the system begins with the collection of dengue case records and climate data from relevant sources. This data is then cleaned and prepared through preprocessing techniques such as normalization and feature selection. The processed data is fed into machine learning models to generate outbreak predictions. These predictions are then interpreted using explainable AI methods before being visualized through dashboards that provide geographic and temporal insights into dengue risk patterns.

4.6 Scalability and Security Considerations

The proposed system is designed to be scalable so that it can handle increasing volumes of data from multiple regions and time periods. Cloud-based data storage and modular system architecture can support future expansion of the system to include additional datasets or predictive models.

Security considerations include ensuring secure storage of collected data and restricting unauthorized access to the system. Data will be anonymized where necessary to protect privacy and comply with ethical data usage guidelines.

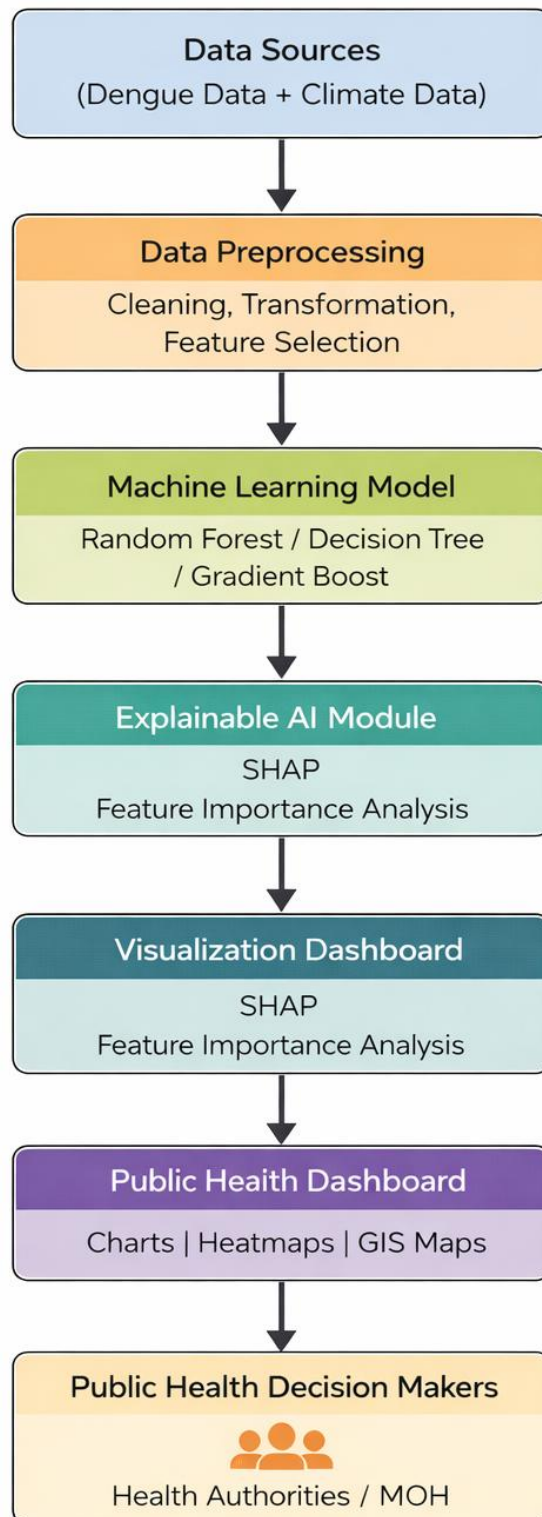


Figure 10 : High-Level Architecture of the Dengue Prediction and Analytics System

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The proposed dengue analytics dashboard will be used by several stakeholders involved in dengue surveillance and public health monitoring.

Public Health Officials from organizations such as the National Dengue Control Unit will use the system to monitor dengue trends, identify high-risk areas, and support outbreak response planning.

Health Researchers and Epidemiologists can use the platform to analyze historical dengue patterns and explore relationships between environmental factors and dengue incidence.

Government Health Authorities may use the dashboard to support decision-making and develop strategies for dengue prevention and control.

System Administrators will manage the platform, maintain the database, and ensure the proper functioning of the system.

5.2 Functional Requirements

The system shall provide the following functionalities:

FR1: The system shall allow users to view dengue case trends over time using interactive charts.

FR2: The system shall visualize dengue case data collected from the National Dengue Control Unit.

FR3: The system shall integrate climate data from the Department of Meteorology to analyze environmental factors related to dengue outbreaks.

FR4: The system shall display predictive outputs generated by the machine learning module.

FR5: The system shall visualize SHAP-based feature importance to explain the factors influencing dengue predictions.

FR6: The system shall provide GIS-based heatmaps to identify high-risk dengue areas within the Colombo district.

FR7: The system shall allow users to filter data based on time period, location, and environmental variables.

FR8: The system shall provide an interactive dashboard interface accessible through web or mobile devices.

5.3 Non-Functional Requirements

Non-functional requirements define the system specifications and quality attributes.

- **Performance:**

The system should load dashboard visualizations within a few seconds to ensure smooth user interaction.

- **Usability:**

The dashboard interface should be user-friendly and easy to understand for public health officials who may not have technical expertise.

- **Security:**

Access to the system should be restricted to authorized users through authentication mechanisms.

- **Scalability:**

The system should be able to handle larger datasets and expand to include other districts in Sri Lanka in the future.

- **Reliability:**

The system should ensure consistent performance and accurate visualization of data.

- **Compatibility:**

The dashboard should be accessible through modern web browsers and mobile devices.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target users of the proposed dengue analytics platform are **public health authorities and organizations responsible for dengue surveillance and disease control**. These include the **National Dengue Control Unit (NDU) of Sri Lanka**, regional health offices, and epidemiology units responsible for monitoring disease outbreaks.

Secondary users include **research institutions, universities, and healthcare researchers** who analyze epidemiological data to study disease patterns and environmental factors influencing dengue outbreaks.

In addition, **government health policymakers** may use the platform to support data-driven decision-making when planning dengue prevention programs and allocating healthcare resources.

6.2 Value Proposition

The proposed system provides a **data-driven decision support platform** that transforms complex dengue surveillance data into clear and actionable insights.

Unlike traditional dengue monitoring systems that rely mainly on historical reports, this platform integrates **data analytics, predictive insights, and interactive visualization dashboards**. These features allow health authorities to identify dengue risk trends, analyze climate correlations, and detect high-risk areas more efficiently.

By improving the interpretability and accessibility of dengue surveillance data, the system can support **early outbreak detection, better resource allocation, and improved public health response strategies**.

6.3 Revenue Model

Since the platform primarily targets public health organizations and research institutions, the system could be implemented as a **government-supported analytical platform** or as a **software-as-a-service (SaaS) solution for health data analytics**.

Revenue could be generated through **institutional licensing, government partnerships, and research collaborations**. Health agencies or organizations could subscribe to the platform to access advanced analytics and visualization features.

6.4 Cost Estimation

The development and deployment of the system will involve several cost components including software development, cloud hosting, and data storage infrastructure.

Initial development costs include software development tools, computing resources, and system implementation efforts. Additional costs may include cloud server hosting, database management, and maintenance of the dashboard platform.

6.5 Pricing Strategy

If the system is deployed as a commercial platform, pricing could follow a **subscription-based model**, where organizations pay a periodic fee for access to analytics dashboards and system updates.

For government agencies or public health organizations, the platform may also be provided through **government-funded projects or research partnerships**, allowing broader adoption without significant financial barriers.

6.6 Competitive Advantage

The proposed platform offers several advantages compared to traditional dengue monitoring systems. The system integrates **data analytics, explainable AI outputs, GIS-based spatial analysis, and interactive dashboards** within a single platform.

This integrated approach improves the usability and interpretability of dengue prediction systems, enabling health officials to quickly understand disease trends and risk patterns. The focus on **visual analytics and explainable insights** distinguishes the system from many existing prediction models that provide limited interpretability.

6.7 Intellectual Property Considerations

The system architecture, dashboard design, and data analytics framework developed in this research could be protected through **software intellectual property rights**. The implementation of the dashboard platform and visualization techniques may be documented and registered as part of the project's intellectual property.

At the same time, the research outputs and analytical methods may also be shared through academic publications to contribute to the broader research community working on public health data analytics.

7. BUDGET AND JUSTIFICATION

The development and implementation of the proposed dengue analytics dashboard require several resources including hardware, software tools, cloud services, and development effort.

The estimated budget for the project is summarized below.

7.1 Estimated Project Budget

Item	Description	Estimated Cost (LKR)
Hardware	Personal computer/laptop for development and testing	6,000
Software Tools	Development tools such as Python libraries and dashboard frameworks	1,500
Cloud Services	Cloud hosting and database services for storing and processing data	4,000
GIS and Visualization Tools	Tools for generating spatial visualizations and maps	1,500
Human Effort	Time spent on system development, testing, and implementation	2,000
Total Estimated Cost		15,000

Table 4: Project Budget Estimation

7.2 Justification of Budget

Hardware:

A computer system with sufficient processing capability is required to perform data preprocessing, analytics, and dashboard development.

Software Tools:

Most of the tools used in the project, such as Python libraries (Pandas, NumPy, Scikit-learn, and SHAP), are open-source. However, additional development tools or visualization plugins may require minor licensing costs.

Cloud Services:

Cloud platforms may be used to host the database and analytics dashboard so that the system can be accessed remotely by users.

GIS and Visualization Tools:

Geospatial tools such as Mapbox or Leaflet will be used to create dengue risk maps and heatmaps.

Human Effort:

This includes the time and effort spent on data analysis, dashboard development, system integration, and testing.

8. WORK BREAKDOWN STRUCTURE (WBS)

The work breakdown structure outlines the major tasks involved in the research project and distributes responsibilities among team members to ensure balanced workload and effective project execution.

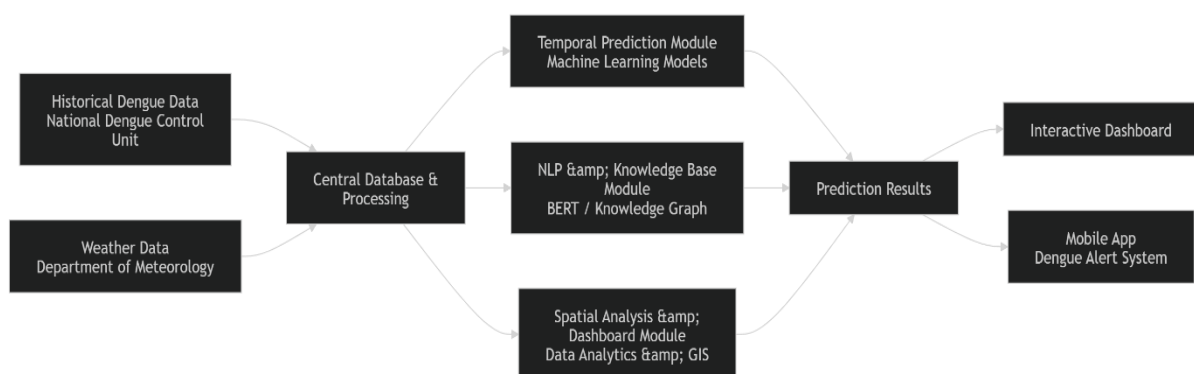


Figure 11 : Work breakdown Structure

8.1 Workload Distribution

The project tasks are distributed among team members according to their respective components to ensure balanced workload and efficient development. Each member focuses on a specific system component while collaborating during integration, testing, and documentation phases.

8.2 Timeline Alignment

The work breakdown structure is aligned with the overall project timeline and milestones defined in the Gantt chart. Each task is scheduled sequentially to ensure smooth development and integration of system components.

9. GANTT CHART

The Gantt chart illustrates the project timeline and key milestones for the development of the dengue analytics dashboard system. The timeline outlines the sequential phases of the research including literature review, data collection, model integration, dashboard implementation, and final evaluation.

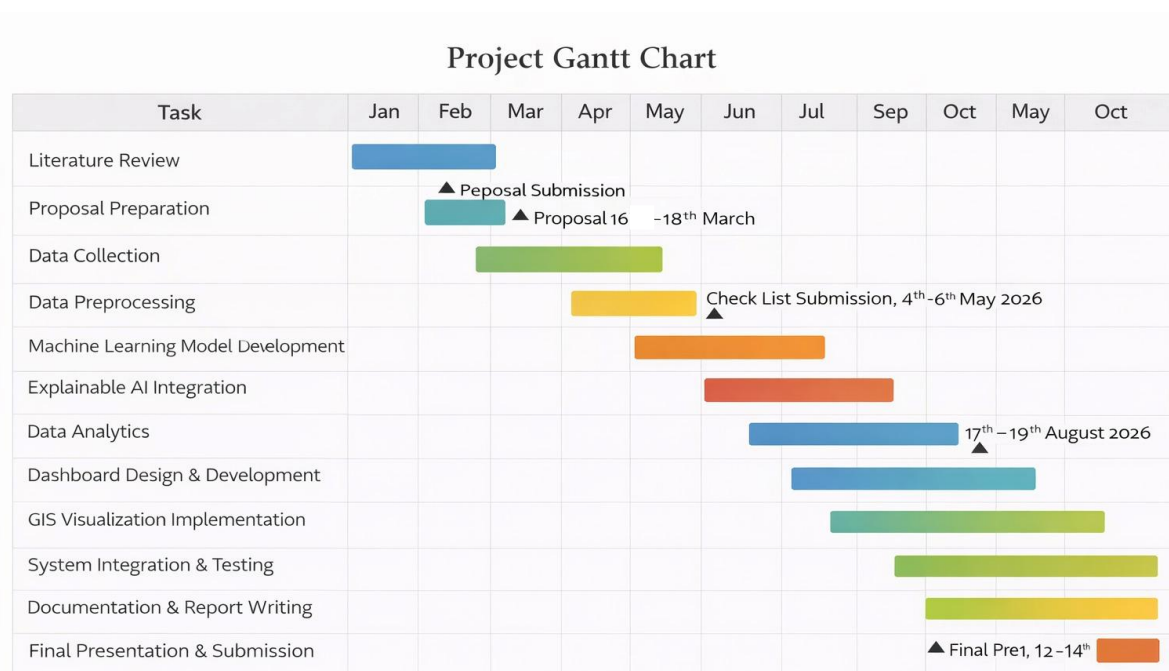


Figure 12 : Project Gantt Chart

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